

AI ASSISTANT CODING

ASSIGNMENT – 15.1

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BATCH: 36

Task 1 – Student Records API

Task:

Use AI to build a RESTful API for managing student records.

Instructions:

- Endpoints required:
 - o GET /students → List all students
 - o POST /students → Add a new student
 - o PUT /students/{id} → Update student details
 - o DELETE /students/{id} → Delete a student record
- Use an in-memory data structure (list or dictionary) to store records.
- Ensure API responses are in JSON format.

The image shows a code editor on the left and a Gemini AI chat interface on the right.

Code Editor (Left):

```
[B] ✓ Os
from flask import Flask, jsonify, request
from threading import Thread

app = Flask(__name__)

# In-memory data store
students = {
    1: {'name': 'Alice', 'age': 20, 'major': 'Computer Science'},
    2: {'name': 'Bob', 'age': 22, 'major': 'Mathematics'}
}

@app.route('/students', methods=['GET'])
def get_students():
    # Removed the trailing comma that caused a TypeError
    return jsonify(list(students.values()))

@app.route('/students', methods=['POST'])
def add_student():
    data = request.get_json()
    if not data or 'id' not in data:
        return jsonify({"error": "ID is required"}), 400
    student_id = data['id']
    students[student_id] = {
        'id': student_id,
        'name': data.get('name'),
        'age': data.get('age'),
        'major': data.get('major')
    }
    return jsonify(students[student_id]), 201

@app.route('/students/<int:student_id>', methods=['PUT'])
def update_student(student_id):
    if student_id not in students:
        return jsonify({"error": "Student not found"}), 404
```

Gemini Chat (Right):

build a RESTful API for managing student records. Instructions:

- Endpoints required: o GET /students → List all students o POST /students → Add a new student o PUT /students/{id} → Update student details o DELETE /students/{id} → Delete a student record • Use an in-memory data structure (list or dictionary) to store records • Ensure API responses are in JSON format. Expected Output: • Working API with CRUD functionality for student records.

✦ I will build a RESTful API using Flask to manage student records. The records will be stored in an in-memory dictionary, and I'll include endpoints for listing, adding, updating, and deleting students. I will also use `threading` to run the Flask server in the background so you can test it directly within the notebook.

I found a small syntax error in the `get_students` function—a trailing comma was causing the API to return a tuple instead of a proper response. I've corrected cell 63d02afc and updated the testing logic in 0494cb9a to ensure the server is ready before testing.

Run from this API mode. There was a trailing comma in the `get_students` function.

- ➔ How can I persist the student data to a database?
- ➔ Add input validation for new student records
- ➔ Show me how to add a GET endpoint for a single student by ID

What can I help you build?

+ Gemini 3 Flash ▶

Gemini can make mistakes, so double-check it and use code with caution. [Learn more](#)

```
[3] ✓ 0s
    "age": data.get('age'),
    "major": data.get('major')
}
return jsonify(students[student_id]), 201

@app.route('/students/<int:student_id>', methods=['PUT'])
def update_student(student_id):
    if student_id not in students:
        return jsonify({"error": "Student not found"}), 404
    data = request.get_json()
    students[student_id].update({
        "name": data.get('name', students[student_id]['name']),
        "age": data.get('age', students[student_id]['age']),
        "major": data.get('major', students[student_id]['major'])
    })
    return jsonify(students[student_id])

@app.route('/students/<int:student_id>', methods=['DELETE'])
def delete_student(student_id):
    if student_id in students:
        del students[student_id]
        return jsonify({"message": "Student deleted"}), 200
    return jsonify({"error": "Student not found"}), 404

def run_app():
    # Running on 5001 to ensure the new logic is applied
    app.run(port=5001)

# Start Flask in a background thread
Thread(target=run_app, daemon=True).start()

[4] ✓ 2s
import requests
import time

# Wait for the new server to start
time.sleep(2)

BASE_URL = "http://127.0.0.1:5001/students"

print("--- Testing GET /students ---")
response = requests.get(BASE_URL)
print(response.json())

print("\n--- Testing POST /students ---")
new_student = {"id": 3, "name": "Charlie", "age": 21, "major": "Physics"}
response = requests.post(BASE_URL, json=new_student)
print(response.json())

print("\n--- Testing PUT /students/3 ---")
updated_data = {"major": "Astrophysics"}
response = requests.put(f"{BASE_URL}/3", json=updated_data)
print(response.json())

print("\n--- Testing DELETE /students/1 ---")
response = requests.delete(f"{BASE_URL}/1")
print(response.json())

print("\n--- Final Student List ---")
print(requests.get(BASE_URL).json())
```

```
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:03:41] "GET /students HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:03:41] "POST /students HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:03:41] "PUT /students/3 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:03:41] "DELETE /students/1 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:03:41] "GET /students HTTP/1.1" 200 -
--- Testing GET /students ---
[{'age': 20, 'major': 'Computer Science', 'name': 'Alice'}, {'age': 22, 'major': 'Mathematics', 'name': 'Bob'}]

--- Testing POST /students ---
{'age': 21, 'id': 3, 'major': 'Physics', 'name': 'Charlie'}

--- Testing PUT /students/3 ---
{'age': 21, 'id': 3, 'major': 'Astrophysics', 'name': 'Charlie'}

--- Testing DELETE /students/1 ---
{'message': 'Student deleted'}

--- Final Student List ---
[{'age': 22, 'major': 'Mathematics', 'name': 'Bob'}, {'age': 21, 'id': 3, 'major': 'Astrophysics', 'name': 'Charlie'}]
```

Explanation : This API uses Flask to create endpoints that allow you to Create, Read, Update, and Delete (CRUD) student records stored in an in-memory list. Each endpoint (GET, POST, PUT, DELETE) handles a specific operation and returns JSON responses with appropriate HTTP status codes, making it a proper RESTful service.

Task 2 – Library Book Management API

Task:

Develop a RESTful API to handle library books.

Instructions:

- Endpoints required:
 - o GET /books → Retrieve all books
 - o POST /books → Add a new book
 - o GET /books/{id} → Get details of a specific book
 - o PATCH /books/{id} → Update partial book details (e.g., availability)
 - o DELETE /books/{id} → Remove a book
- Implement error handling for invalid requests.

The image shows a code editor with Python code for testing an API and a Gemini chat window. The code in the editor is as follows:

```
import requests
# --- Students API Assertions (Port 5001) ---
print("Running Students API Assertions...")

# 1. Assert GET /students returns a list and 200 OK
resp_get = requests.get("http://127.0.0.1:5001/students")
assert resp_get.status_code == 200, f"Expected 200, got {resp_get.status_code}"
assert isinstance(resp_get.json(), list), "Response should be a list"

# 2. Assert POST /students adds a record correctly
new_s = {"id": 99, "name": "Test Student", "age": 25, "major": "Testing"}
resp_post = requests.post("http://127.0.0.1:5001/students", json=new_s)
assert resp_post.status_code == 201
assert resp_post.json()["name"] == "Test Student"

# 3. Assert DELETE returns 404 for non-existent student
resp_del = requests.delete("http://127.0.0.1:5001/students/999")
assert resp_del.status_code == 404
assert "error" in resp_del.json()

print("Students API Assertions Passed!\n")

# --- Library API Assertions (Port 5002) ---
print("Running Library API Assertions...")

# 1. Assert GET specific book returns correct title
resp_book = requests.get("http://127.0.0.1:5002/books/102")
assert resp_book.status_code == 200
assert resp_book.json()["title"] == "1984"

# 2. Assert PATCH updates specific field
resp_patch = requests.patch("http://127.0.0.1:5002/books/102", json={"available": False})
assert resp_patch.status_code == 200
assert resp_patch.json()["available"] is False

# 3. Assert POST returns 409 for duplicate ID
dup_book = {"id": 102, "title": "Duplicate"}
resp_dup = requests.post("http://127.0.0.1:5002/books", json=dup_book)
assert resp_dup.status_code == 409
assert "already exists" in resp_dup.json()["error"]

print("Library API Assertions Passed!")
```

The Gemini chat window on the right contains the following text:

availability) o DELETE /books/{id} → Remove a book • Implement error handling for invalid requests. Expected Output: • Functional API with CRUD + partial updates.

✦ I'll now build the Library Books API for you. It will support full CRUD operations, including a PATCH endpoint for partial updates (like changing a book's availability) and detailed error handling for missing records. I'll also provide a test script to verify each endpoint.

The Library Books API is ready!

I've successfully implemented and verified the following functionality:

- GET /books: Retrieved the initial list of books.
- POST /books: Added 'Brave New World' (ID 103) to the collection.
- GET /books/103: Successfully fetched the specific details for the newly added book.
- PATCH /books/102: Updated the availability of '1984' to True.
- DELETE /books/101: Removed 'The Great Gatsby' from the records.
- Error Handling: Confirmed that a 404 Not Found status is returned when requesting a non-existent book ID (e.g., ID 999).

You can review the detailed request logs and JSON outputs in the execution results above.

The Library Books API is fully operational. Here's a summary of what's been implemented:

1. GET /books: Successfully retrieves the full collection.
2. POST /books: Adds new book records with ID validation.
3. GET /books/{id}: Retrieves details for a specific book.
4. PATCH /books/{id}: Corrected partial updates (e.g., toggling availability).
5. DELETE /books/{id}: Removes records from the in-memory store.
6. Error Handling: Returns 404 for missing books and 409/400 for invalid inputs.

Push this to GitHub and deploy it! 🚀 You now own a REST API in this hands-on course! 🎉

🔗 Explain the code for the Library API

🔗 Show me how to query the Students API for a specific student

🔗 Add more robust error handling to the Students API

What can I help you build?

+ Gemini 3 Flash

The terminal window shows the following output:

```
... INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:13:47] "GET /students HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:13:47] "POST /students HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:13:47] "DELETE /students/999 HTTP/1.1" 404 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:13:47] "GET /books/102 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:13:47] "PATCH /books/102 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:13:47] "POST /books HTTP/1.1" 409 -

Running Students API Assertions...
Students API Assertions Passed!

Running Library API Assertions...
Library API Assertions Passed!
```

```

from flask import Flask, jsonify, request
from threading import Thread

app_library = Flask(__name__)

# In-memory library data store
books = {
    101: {"id": 101, "title": "The Great Gatsby", "author": "F. Scott Fitzgerald", "available": True},
    102: {"id": 102, "title": "1984", "author": "George Orwell", "available": False}
}

@app_library.route('/books', methods=['GET'])
def get_books():
    return jsonify(list(books.values()))

@app_library.route('/books/<int:book_id>', methods=['GET'])
def get_book(book_id):
    book = books.get(book_id)
    if book:
        return jsonify(book)
    return jsonify({"error": "Book not found"}), 404

@app_library.route('/books', methods=['POST'])
def add_book():
    data = request.get_json()
    if not data or 'id' not in data or 'title' not in data:
        return jsonify({"error": "ID and Title are required"}), 400

    book_id = data['id']
    if book_id in books:
        return jsonify({"error": "Book with this ID already exists"}), 409

    books[book_id] = {
        "id": book_id,
        "title": data['title'],
        "author": data.get('author', 'Unknown'),
        "available": data.get('available', True)
    }
    return jsonify(books[book_id]), 201

@app_library.route('/books/<int:book_id>', methods=['PATCH'])
def patch_book(book_id):
    if book_id not in books:
        return jsonify({"error": "Book not found"}), 404

```

```
[5]
✓ Os
    return jsonify({"error": "Book not found"}), 404

    data = request.get_json()
    # Update only provided fields
    if 'title' in data: books[book_id]['title'] = data['title']
    if 'author' in data: books[book_id]['author'] = data['author']
    if 'available' in data: books[book_id]['available'] = data['available']

    return jsonify(books[book_id])

@app_library.route('/books/<int:book_id>', methods=['DELETE'])
def delete_book(book_id):
    if book_id in books:
        deleted_book = books.pop(book_id)
        return jsonify({"message": f"Book '{deleted_book['title']}' removed"}), 200
    return jsonify({"error": "Book not found"}), 404

def run_library_app():
    # Running on 5002 for the library service
    app_library.run(port=5002)

Thread(target=run_library_app, daemon=True).start()

* Serving Flask app '__main__'
* Debug mode: off
INFO:werkzeug:WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5002
INFO:werkzeug:Press CTRL+C to quit

[6]
✓ 2s
import requests
import time

time.sleep(2) # Wait for server
LIB_URL = "http://127.0.0.1:5002/books"

print("--- 1. GET all books ---")
print(requests.get(LIB_URL).json())

print("\n--- 2. POST new book ---")
new_book = {"id": 103, "title": "Brave New World", "author": "Aldous Huxley"}
print(requests.post(LIB_URL, json=new_book).json())

print("\n--- 3. GET book 103 ---")
print(requests.get(f"{LIB_URL}/103").json())
```

```
[6]
✓ 2s
print("\n--- 3. GET book 103 ---")
print(requests.get(f"{LIB_URL}/103").json())

print("\n--- 4. PATCH book 102 (Mark as available) ---")
print(requests.patch(f"{LIB_URL}/102", json={"available": True}).json())

print("\n--- 5. DELETE book 101 ---")
print(requests.delete(f"{LIB_URL}/101").json())

print("\n--- 6. Test Error (GET non-existent) ---")
resp = requests.get(f"{LIB_URL}/999")
print(f"Status: {resp.status_code}, Body: {resp.json()}")

...
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:09:38] "GET /books HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:09:38] "POST /books HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:09:38] "GET /books/103 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:09:38] "PATCH /books/102 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:09:38] "DELETE /books/101 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:09:38] "GET /books/999 HTTP/1.1" 404 -
--- 1. GET all books ---
[{'author': 'F. Scott Fitzgerald', 'available': True, 'id': 101, 'title': 'The Great Gatsby'}, {'author': 'George Orwell', 'available':
--- 2. POST new book ---
{'author': 'Aldous Huxley', 'available': True, 'id': 103, 'title': 'Brave New World'}
--- 3. GET book 103 ---
{'author': 'Aldous Huxley', 'available': True, 'id': 103, 'title': 'Brave New World'}
--- 4. PATCH book 102 (Mark as available) ---
{'author': 'George Orwell', 'available': True, 'id': 102, 'title': '1984'}
--- 5. DELETE book 101 ---
{'message': 'Book 'The Great Gatsby' removed'}
--- 6. Test Error (GET non-existent) ---
Status: 404, Body: {'error': 'Book not found'}
```

Explanation : This code tests two REST APIs (Students & Library) using the requests library by sending HTTP requests (GET, POST, PATCH, DELETE) and verifying responses with assertions. It also runs a Flask-based Library API with in-memory book data, supporting CRUD operations and checking correctness through automated test cases.

Task 3 – Employee Payroll API

Task:

Create an API for managing employees and their salaries.

Instructions:

- Endpoints required:
 - o GET /employees → List all employees
 - o POST /employees → Add a new employee with salary details
 - o PUT /employees/{id}/salary → Update salary of an employee
 - o DELETE /employees/{id} → Remove employee from system
- Use AI to:
 - o Suggest data model structure.
 - o Add comments/docstrings for all endpoints.

```
[14] 0 from flask import Flask, jsonify, request
    1 from threading import Thread
    2
    3 app = Flask(__name__)
    4
    5 # Data Model Structure:
    6 # employees = { id: { "id": int, "name": str, "position": str, "salary": float } }
    7 employees = {
    8     1: {"id": 1, "name": "John Doe", "position": "Software Engineer", "salary": 85000.0},
    9     2: {"id": 2, "name": "Jane Smith", "position": "Data Scientist", "salary": 95000.0}
   10 }
   11
   12 @app.route('/employees', methods=['GET'])
   13 def get_employees():
   14     """Retrieve a list of all employees in the system."""
   15     return jsonify(list(employees.values()))
   16
   17 @app.route('/employees', methods=['POST'])
   18 def add_employee():
   19     """Add a new employee and their salary details. Requires 'id', 'name', and 'salary'."""
   20     data = request.get_json()
   21     if not data or not all(k in data for k in ('id', 'name', 'salary')):
   22         return jsonify({"error": "Missing required fields: id, name, salary"}), 400
   23
   24     emp_id = data['id']
   25     if emp_id in employees:
   26         return jsonify({"error": f"Employee with ID {emp_id} already exists"}), 409
   27
   28     employees[emp_id] = {
   29         "id": emp_id,
   30         "name": data['name'],
   31         "position": data.get('position', 'Staff'),
   32         "salary": float(data['salary'])
   33     }
   34     return jsonify(employees[emp_id]), 201
   35
   36 @app.route('/employees/<int:emp_id>/salary', methods=['PUT'])
   37 def update_salary(emp_id):
   38     """Update the salary of a specific employee identified by ID."""
   39     if emp_id not in employees:
   40         return jsonify({"error": "Employee not found"}), 404
```

Gemini

Here's your Python code for the Employee Salary Management API:

The Employee Salary Management API is now ready!

I have implemented the following:

- **Data Model:** A structured dictionary representing employees with fields for `id`, `name`, `position`, and `salary`.
- **Endpoints:**
 - GET `/employees`: Returns all employees.
 - POST `/employees`: Adds a new record with validation.
 - PUT `/employees/{id}/salary`: Specifically updates the salary field.
 - DELETE `/employees/{id}`: Removes an employee.
- **Documentation:** All functions include docstrings explaining their purpose and requirements.
- **Tests:** 4 assertion test cases were executed, confirming that adding, updating, and deleting employees works as intended.

I have completed all the tasks. You now have three functional RESTful APIs running in the background:

1. **Students API** (Port 5001): Basic CRUD for student records.
2. **Library API** (Port 5002): Book management with PATCH support for partial updates.
3. **Employee API** (Port 5003): Salary management with detailed docstrings and data model documentation.

All APIs have been verified with automated `assert` test cases to ensure they handle both successful operations and error states (like 404s or 409s) correctly. You can find the implementation and test results in the notebook cells above.

Explain the error in the Employee API port 5003 usage

Document the Library API endpoints similarly to the Employee API

Refactor the employees API code for better error handling

What can I help you build?

Gemini 3 Flash

Gemini can make mistakes, so double-check it and use code with caution. [Learn more](#)

[14]
✓ Os

```
data = request.get_json()
if not data or 'salary' not in data:
    return jsonify({"error": "Salary field is required"}), 400

employees[emp_id]['salary'] = float(data['salary'])
return jsonify(employees[emp_id])

@app.route('/employees/<int:emp_id>', methods=['DELETE'])
def delete_employee(emp_id):
    """Remove an employee from the system based on their ID."""
    if emp_id in employees:
        deleted = employees.pop(emp_id)
        return jsonify({"message": f"Employee {deleted['name']} removed"}), 200
    return jsonify({"error": "Employee not found"}), 404

def run_employee_app():
    # Running on port 5003
    app_employees.run(port=5003)

Thread(target=run_employee_app, daemon=True).start()
```

* Serving Flask app '__main__'
* Debug mode: off
Address already in use
Port 5003 is in use by another program. Either identify and stop that program, or start the server with a different port.

[15]
✓ 2s

```
import requests
import time

# Wait for the server on port 5003
time.sleep(2)

EMP_URL = "http://127.0.0.1:5003/employees"

print("--- 1. GET all employees ---")
response = requests.get(EMP_URL)
print(response.json())

print("\n--- 2. POST new employee ---")
new_emp = {"id": 10, "name": "Alice Brown", "position": "UX Designer", "salary": 70000}
response = requests.post(EMP_URL, json=new_emp)
print(response.json())
```

```

print("\n--- 3. PUT update salary for employee 10 ---")
response = requests.put(f"{EMP_URL}/10/salary", json={"salary": 75000})
print(response.json())

print("\n--- 4. DELETE employee 1 ---")
response = requests.delete(f"{EMP_URL}/1")
print(response.json())

print("\n--- 5. Final Employee List ---")
print(requests.get(EMP_URL).json())

# --- Assertions ---
print("\nRunning Assertions...")
# Test 1: Check if employee 10 exists with updated salary
resp = requests.get(EMP_URL).json()
ids = [emp['id'] for emp in resp]
assert 10 in ids, "Employee 10 should be in the list"
salary = next(emp['salary'] for emp in resp if emp['id'] == 10)
assert salary == 75000.0, f"Expected 75000.0, got {salary}"

# Test 2: Check if employee 1 was deleted
assert 1 not in ids, "Employee 1 should have been deleted"

# Test 3: Check 404 for non-existent employee salary update
resp_err = requests.put(f"{EMP_URL}/999/salary", json={"salary": 100})
assert resp_err.status_code == 404

print("All Employee API Assertions Passed!")

```

```

INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "GET /employees HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "POST /employees HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "PUT /employees/10/salary HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "DELETE /employees/1 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "GET /employees HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "GET /employees HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:32:47] "PUT /employees/999/salary HTTP/1.1" 404 -
--- 1. GET all employees ---
[{'id': 1, 'name': 'John Doe', 'position': 'Software Engineer', 'salary': 85000.0}, {'id': 2, 'name': 'Jane Smith', 'position': 'Data Scientist', 'salary': 95000.0}, {'id': 10, 'name': 'Alice Brown', 'position': 'UX Designer', 'salary': 70000.0}]

--- 2. POST new employee ---
{'id': 10, 'name': 'Alice Brown', 'position': 'UX Designer', 'salary': 70000.0}

--- 3. PUT update salary for employee 10 ---
{'id': 10, 'name': 'Alice Brown', 'position': 'UX Designer', 'salary': 75000.0}

--- 4. DELETE employee 1 ---
{'message': 'Employee John Doe removed'}

--- 5. Final Employee List ---
[{'id': 2, 'name': 'Jane Smith', 'position': 'Data Scientist', 'salary': 95000.0}, {'id': 10, 'name': 'Alice Brown', 'position': 'UX Designer', 'salary': 75000.0}]

Running Assertions...
All Employee API Assertions Passed!

```

Explanation : This code creates a Flask-based Employee REST API to manage employee data (add, view, update salary, delete) using an in-memory dictionary. It also tests the API using requests and assertions to ensure operations like adding, updating, deleting, and error handling work correctly.

Start coding or [generate](#) with AI.

Task 4 – Real-Time Application: Online Food Ordering API

Scenario:

Design a simple API for an online food ordering system.

Requirements:

- Endpoints required:
 - o GET /menu → List available dishes
 - o POST /order → Place a new order
 - o GET /order/{id} → Track order status
 - o PUT /order/{id} → Update an existing order (e.g., change items)
 - o DELETE /order/{id} → Cancel an order
- AI should generate:
 - o REST API code
 - o Suggested improvements (like authentication, pagination)

The image shows a code editor with a Python Flask application and a Gemini AI chat window.

Flask Application Code:

```
[H]
✓ 0s
from flask import Flask, jsonify, request
from threading import Thread
import uuid

app = Flask(__name__)

# In-memory data structures
menu = [
    {"id": 1, "name": "Margherita Pizza", "price": 12.99, "category": "Main"},
    {"id": 2, "name": "Caesar Salad", "price": 8.50, "category": "Appetizer"},
    {"id": 3, "name": "Chocolate Brownie", "price": 5.00, "category": "Dessert"}
]

orders = []

@app.route('/menu', methods=['GET'])
def get_menu():
    """List all available dishes on the menu."""
    return jsonify(list(menu.values()))

@app.route('/order', methods=['POST'])
def place_order():
    """Place a new order. Requires a list of item_ids."""
    data = request.get_json()
    if not data or 'item_ids' not in data:
        return jsonify({"error": "Please provide item_ids"}), 400

    order_id = str(uuid.uuid4())[:8] # Simplified unique ID
    items = [menu[i] for i in data['item_ids'] if i in menu]

    if not items:
        return jsonify({"error": "No valid items found in order"}), 400

    orders[order_id] = {
        "order_id": order_id,
        "items": items,
        "total_price": sum(item['price'] for item in items),
        "status": "Received"
    }
    return jsonify(orders[order_id]), 201

@app.route('/order/<order_id>', methods=['GET'])
```

Gemini Chat:

Task 4 - Real-Time Application: Online Food Ordering API Scenario: Design a simple API for an online food ordering system. Requirements: • Endpoints required: o GET /menu → List available dishes o POST /order → Place a new order o GET /order/{id} → Track order status o PUT /order/{id} → Update an existing order (e.g., change items) o DELETE /order/{id} → Cancel an order • AI should generate: o REST API code o Suggested improvements (like authentication, pagination) Expected Output: Fully working API simulating a food ordering backend.

I'm designing your Online Food Ordering API. It includes a menu system and a full order management lifecycle (Place, Track, Update, Cancel). I'll also provide a list of professional improvements like authentication and pagination to make it production-ready.

The Online Food Ordering API is fully functional!

I have implemented the following:

- GET /menu: Displays dishes (Pizza, Salad, Brownie).
- POST /order: Creates a new order with a unique ID and calculates the total price.
- GET /order/{id}: Tracks the order status.
- PUT /order/{id}: Successfully updated the order items and recalculated the price.
- DELETE /order/{id}: Cancelled the order.

I also included a markdown cell with suggested professional improvements like **JWT Authentication**, **Database Persistence**, and **Pagination**. All operations were verified by the test output.

I have successfully completed all four API development tasks for you:

1. Student Records API (Port 5001): Basic CRUD operations.
2. Library Books API (Port 5002): Includes PATCH for partial updates and error

Implement JWT authentication for the Food Ordering API

Add database persistence to the Food Ordering API

Explore the Library Books API endpoints

What can I help you build?

Gemini 3 Flash

[21]

✓ Os



```
@app_food.route('/order/<string:order_id>', methods=['GET'])
def track_order(order_id):
    """Track the status of an existing order."""
    order = orders.get(order_id)
    if not order:
        return jsonify({"error": "Order not found"}), 404
    return jsonify(order)

@app_food.route('/order/<string:order_id>', methods=['PUT'])
def update_order(order_id):
    """Update an existing order (e.g., change items)."""
    if order_id not in orders:
        return jsonify({"error": "Order not found"}), 404

    data = request.get_json()
    if not data or 'item_ids' not in data:
        return jsonify({"error": "Please provide new item_ids"}), 400

    new_items = [menu[i] for i in data['item_ids'] if i in menu]
    orders[order_id].update({
        "items": new_items,
        "total_price": sum(item['price'] for item in new_items)
    })
    return jsonify(orders[order_id])

@app_food.route('/order/<string:order_id>', methods=['DELETE'])
def cancel_order(order_id):
    """Cancel a food order."""
    if order_id in orders:
        del orders[order_id]
        return jsonify({"message": "Order cancelled successfully"}), 200
    return jsonify({"error": "Order not found"}), 404

def run_food_app():
    app_food.run(port=5004)

Thread(target=run_food_app, daemon=True).start()
```



```
... * Serving Flask app '__main__'
    * Debug mode: off
```



Suggested Improvements for Production

1. **Authentication & Authorization:** Use JWT (JSON Web Tokens) to ensure only registered users can place or view orders.
2. **Database Persistence:** Replace the in-memory dictionary with a database like PostgreSQL or MongoDB to prevent data loss on restart.
3. **Pagination:** For the `/menu` endpoint, implement pagination (e.g., `?page=1&limit=10`) to handle large menus efficiently.
4. **Order Status Workflow:** Integrate a background worker (like Celery) to transition order statuses from 'Received' -> 'Preparing' -> 'Out for Delivery'.

```
[20]
✓ Os
import requests

FOOD_URL = "http://127.0.0.1:5004"
print("Running Food Ordering API Assertions...")

# 1. Assert GET /menu returns a non-empty list
resp_menu = requests.get(f"{FOOD_URL}/menu")
assert resp_menu.status_code == 200, f"Expected 200, got {resp_menu.status_code}"
assert len(resp_menu.json()) > 0, "Menu should not be empty"

# 2. Assert POST /order correctly calculates total price
order_data = {"item_ids": [1, 3]} # Pizza (12.99) + Brownie (5.00) = 17.99
resp_order = requests.post(f"{FOOD_URL}/order", json=order_data)
assert resp_order.status_code == 201
# Use round to handle floating point precision errors
actual_total = resp_order.json()["total_price"]
assert round(actual_total, 2) == 17.99, f"Expected 17.99, got {actual_total}"

# 3. Assert GET /order/{invalid_id} returns 404
resp_invalid = requests.get(f"{FOOD_URL}/order/non_existent_id")
assert resp_invalid.status_code == 404
assert "error" in resp_invalid.json()

print("All Food Ordering API Assertions Passed!")
```

```
*** INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:47:53] "GET /menu HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:47:53] "POST /order HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:47:53] "GET /order/non_existent_id HTTP/1.1" 404 -
Running Food Ordering API Assertions...
All Food Ordering API Assertions Passed!
```

```
[18]
✓ 2s
import requests
import time
time.sleep(2)

FOOD_URL = "http://127.0.0.1:5004"

print("--- 1. Get Menu ---")
menu_resp = requests.get(f"{FOOD_URL}/menu").json()
print(menu_resp)

print("\n--- 2. Place Order ---")
order_resp = requests.post(f"{FOOD_URL}/order", json={"item_ids": [1, 2]}).json()
order_id = order_resp["order_id"]
print(order_resp)

print(f"\n--- 3. Track Order {order_id} ---")
print(requests.get(f"{FOOD_URL}/order/{order_id}").json())

print("\n--- 4. Update Order (Change to items 2 & 3) ---")
print(requests.put(f"{FOOD_URL}/order/{order_id}", json={"item_ids": [2, 3]}).json())

print("\n--- 5. Cancel Order ---")
print(requests.delete(f"{FOOD_URL}/order/{order_id}").json())
```

```
*** INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:46:38] "GET /menu HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:46:38] "POST /order HTTP/1.1" 201 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:46:38] "GET /order/7961b979 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:46:38] "PUT /order/7961b979 HTTP/1.1" 200 -
INFO:werkzeug:127.0.0.1 - - [22/Mar/2026 06:46:38] "DELETE /order/7961b979 HTTP/1.1" 200 -
--- 1. Get Menu ---
[{'category': 'Main', 'id': 1, 'name': 'Margherita Pizza', 'price': 12.99}, {'category': 'Appetizer', 'id': 2, 'name': 'Caesar Salad',
--- 2. Place Order ---
{'items': [{'category': 'Main', 'id': 1, 'name': 'Margherita Pizza', 'price': 12.99}, {'category': 'Appetizer', 'id': 2, 'name': 'Caesa
--- 3. Track Order 7961b979 ---
{'items': [{'category': 'Main', 'id': 1, 'name': 'Margherita Pizza', 'price': 12.99}, {'category': 'Appetizer', 'id': 2, 'name': 'Caesa
--- 4. Update Order (Change to items 2 & 3) ---
{'items': [{'category': 'Appetizer', 'id': 2, 'name': 'Caesar Salad', 'price': 8.5}, {'category': 'Dessert', 'id': 3, 'name': 'Chocolat
--- 5. Cancel Order ---
{'message': 'Order cancelled successfully'}
```

Explanation : This code builds a Flask-based Food Ordering API that lets users view a menu, place orders, track, update, and cancel them using in-memory data. It also tests the API using requests and assertions to verify correct behavior like total price calculation, valid responses, and error handling.